

Almanac 47 for the NumWorks

Python version: the ALMANAC page and the horizon chart — user manual

Two graphic views of the C47 suite, drawn with the NumWorks's own graphics module (kandinsky, keys with ion): the “vintage” ALMANAC page and the horizon chart. The numbers come from nav.py and navdata.py — the same series and methods as the C47 programs (the Sun from the full VSOP87 series, valid 2000–2050, nutation, the 58 navigation stars, sunrise, twilight, Moon phase). On a PC every value on the screens was compared with the text page of nav.py and with the C47 ALMANAC screen: identical.

The Sun and the stars only: nav.py has no Moon position and no planets (the Moon phase and age are shown). The table has the Sun and the brightest stars higher than 10°, as on the C47.

It supports, and does not replace, the Nautical Almanac. Always cross-check the values.

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1. Files

File	From	What it is
nav.py, navdata.py	python/	the almanac computation and data (also a text menu)
nwlib.py	python/numworks/	text (the C47 5×7 font), symbols, keys, main loop
almview.py	python/numworks/	starts on the ALMANAC page
skyview.py	python/numworks/	starts on the horizon chart

2. Copy to the calculator

- **Epsilon (official):** on my.numworks.com create each script (Python → new script, paste the file, same name), then “Send to my calculator” with the calculator on USB (Chrome or Edge). The online simulator there runs them too.
- **Upsilon / Omega:** the same five files, with their web installer's script manager or the workshop.
- Open almview.py (or skyview.py) in the Python app and choose **Execute script**.

3. Inputs

The program asks in the terminal, like the nav.py menu:

Input	Format	Example
Year, Month, Day	numbers	2026, 9, 26
UT	hh:mm or hh:mm:ss	14:57
Lat, Lon	“25 20” = 25° 20′, “25.5” = decimal degrees, “-” for S / W	25 20, 55 12

- **Time is UT (UT1):** for a sight timed in UTC add DUT1 (under 0.9 s) — as in the almanac.
- **Signs:** latitude N +, S −; longitude E +, W −.

4. Keys

Key	Action
UP / DOWN	one hour later / earlier
OK or EXE	switch ALMANAC ↔ HORIZON
LEFT / RIGHT	HORIZON: the previous / next body for the top line (shown in red)
BACK	exit

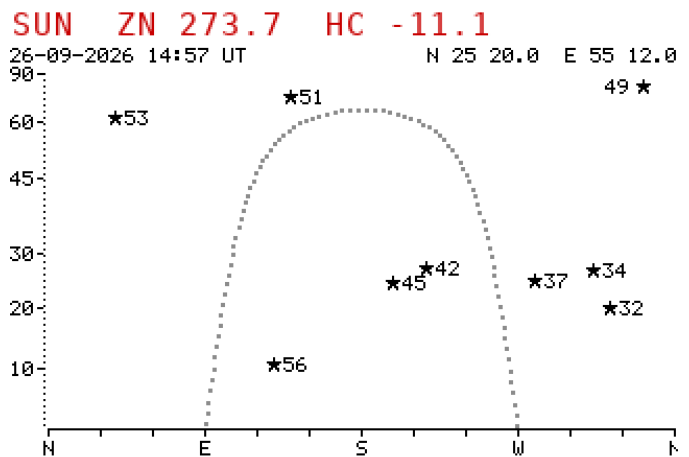
5. The views

The previews were made on a PC with stand-ins for the kandinsky module (the same scripts; the calculator's own font may make text a little wider or narrower).

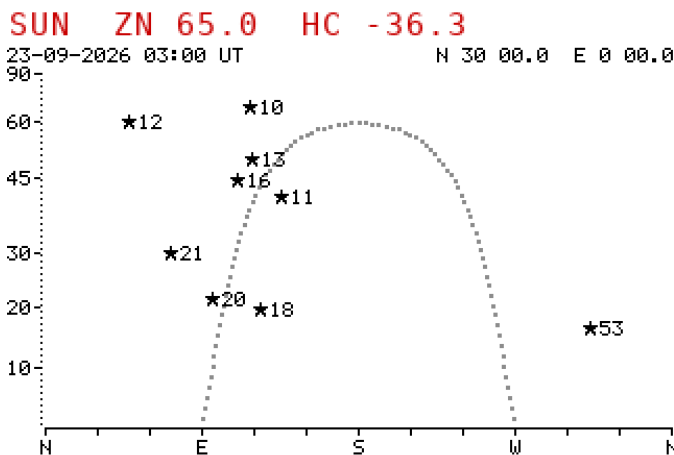
```
26-09-2026 14:57 UT      DR N 25 20.0
                          E 55 12.0
BODY      GHA      DEC      HC      ZN
ARIES      229 40.8
☉ SUN      46 25.7 S 1 24.5 -11 06.4 273.7
★37 ARCTURUS 15 27.7 N 19 02.7 24 59.7 280.2
★49 VEGA    310 13.1 N 38 48.8 75 46.1 342.6
★51 ALTAIR  291 39.3 N 8 56.5 69 23.0 140.4
★42 ANTARES 341 55.1 S 26 29.5 27 00.7 217.3
★56 FOMALHAUT 244 53.4 S 29 28.7 10 35.9 130.0
★53 DENEBO  279 05.4 N 45 22.8 61 13.8 039.3
★45 SHAULA  325 49.4 S 37 07.5 24 29.0 198.3
★32 ALIOTH  35 53.1 N 55 48.9 20 08.5 323.2
NAUT TWI 01:20 15:01      MOON 100% FULL
RISE/SET 02:09 14:12      AGE 15.5 DAYS
MER PASS 08:11 SD 16.0

UP/DOWN 1 HOUR  OK HORIZON  BACK EXIT
DOES NOT REPLACE THE NAUTICAL ALMANAC
```

ALMANAC: date, UT, DR; GHA Aries; BODY, GHA, DEC, HC, ZN of the Sun and the stars; naut. twilight, rise/set, meridian passage, Sun SD, Moon phase and age. A negative Hc is white on black. 26 Sep 2026 14:57 UT, 25° 20' N 055° 12' E.



HORIZON: Hc up (sine scale), Zn across, the celestial equator dotted, the Sun and the stars with their numbers; the top line shows ZN and HC of the selected body. Same time and place.



HORIZON at night: 23 Sep 2026 03:00 UT, 30° N 0° E (the Sun below the horizon).

```
26-09-2026 14:57 UT      DR N 25 20.0
                          E 55 12.0
BODY      GHA      DEC      HC      ZN
ARIES      229 40.8
☉ SUN      46 25.7 S 1 24.5 -11 06.4 273.7
★37 ARCTURUS 15 27.7 N 19 02.7 24 59.7 280.2
★49 VEGA    310 13.1 N 38 48.8 75 46.1 342.6
★51 ALTAIR  291 39.3 N 8 56.5 69 23.0 140.4
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★45 SHAULA  325 49.4 S 37 07.5 24 29.0 198.3
★32 ALIOTH  35 53.1 N 55 48.9 20 08.5 323.2
NAUT TWI 01:20 15:01      MOON 100% FULL
RISE/SET 02:09 14:12      AGE 15.5 DAYS
MER PASS 08:11 SD 16.0

UP/DOWN 1 HOUR  OK HORIZON  BACK EXIT
DOES NOT REPLACE THE NAUTICAL ALMANAC
```

6. The text menu

nav.py alone (Execute script, or **menu()** in the shell) is the text version: 1 Sun, 2 Star, 3 Rise/Set, 4 Moon phase, 5 Almanac (the same page as the ALMANAC view, as text).

7. Memory and size

- The five files are about 26 KB. The star data of navdata.py take about 15 KB of the Python heap.
- Epsilon 19 or later (64 KB heap) or Upsilon / Omega (about 100 KB): fine. Epsilon before 19 (32 KB heap): close to the limit — if you get MemoryError, update Epsilon or use Upsilon, and do not import other scripts in the same session.

8. Limitations

- **Series only, no almanac tables.** The Sun comes from the truncated VSOP87 series: valid 2000–2050, better than 0.1'. The tables TBL_1 / TBL_5 of the C47 and Free42 are not read (they mainly serve the Moon and the planets, which this version does not compute).
- **No Moon position and no planets:** only the Moon phase and age. For the Moon and the planets use the C47, Free42 / DM42n (NAVFULL) or the Python version on the PC (python/native).
- **ΔT fixed** at 69.2 s (TT – UT1), as in the C47 series; the input time is UT1 (add DUT1 to UTC).

- **Two views only:** ALMANAC and HORIZON (no split sky, all-sky, animation or star pages).
- **Memory:** on Epsilon before 19 (32 KB heap) the program is close to the limit (section 7); the almanac tables (1 year is about 8,400 numbers) could not fit in any NumWorks Python heap.

9. Status

- Not yet tried on a real NumWorks. Checked on a PC (stand-in module) and in a MicroPython interpreter.